

Applied Statistics for Engineers and Scientists II

Project-Based Quiz 1: Ridge, Lasso, and Regularization for Neural Features

HCMUS Advanced Program in Computer Science

One-week take-home group project

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Project brief

Aim

Regularization is one of the clearest links between statistical learning and modern machine learning. In this project, your group will study that link at three levels:

1. ordinary least squares, ridge regression, and lasso on an applied regression dataset;
2. the optimization mechanisms behind shrinkage, conditioning, and sparsity;
3. elastic-net regularization on a fixed ReLU feature representation, used as a controlled bridge to neural-network methodology.

The project emphasizes reasoning, reproducibility, and defensible model comparison. A strong report explains not only which method performed well, but why its behavior is consistent with the objective function and the data.

Ridge regression stabilizes estimation through an L_2 penalty (Hoerl and Kennard 1970), while lasso uses an L_1 penalty that can produce sparse coefficient vectors (Tibshirani 1996).

Format and scoring

- Work in groups of **3–5 students**, unless another arrangement is approved.
- The group report contributes **9.0 points**.
- The individual oral defense contributes **1.0 point**.
- Each student’s final score is the group report score plus that student’s oral-defense score, for a maximum of **10.0 points**.

Dataset assignment

Use the course-provided data in `data/`.

- Odd-numbered groups: use `fat.csv` and predict `brozek`.
- Even-numbered groups: use `prostate.csv` and predict `lpsa`.

The body-fat data follow the study described by Johnson (1996), and the prostate data follow the radical-prostatectomy study of Stamey et al. (1989). For `fat.csv`, exclude `siri`, `density`, and `free` from the candidate predictors. These variables encode alternative body-fat calculations or quantities derived using body-fat information and would make the prediction task misleading.

For `prostate.csv`, use all variables except `lpsa` as candidate predictors unless you document a statistical reason for an additional exclusion.

Required software

Use R and declare every package in the report. The supplied template uses `tidyverse`, `glmnet`, `broom`, and `knitr`. Other packages are allowed when they are declared, justified, and recorded by `sessionInfo()`.

Submission and authorship

Files

Submit one zip archive named

`ASESII_24A02_ProjectQuiz1_GroupXX.zip`

containing:

1. `GroupXX_ProjectQuiz1.Rmd`;
2. `GroupXX_ProjectQuiz1.pdf` or `GroupXX_ProjectQuiz1.html`;
3. `README.txt` with exact reproduction instructions;
4. `references.bib` containing every cited source;
5. any helper scripts or external figure files required by the report.

Reproducibility

The report must run from a clean R session with relative paths. Set and report seeds for the train/test split, cross-validation folds, random features, and any repeated analysis. Do not depend on objects stored in the global workspace or `.Rhistory`.

AI use

AI tools may be used for explanation, code review, debugging, or writing support. The group remains responsible for every statement, citation, result, and line of code.

However, you must carefully check the outputs of AI since their arguments may be weak and unreliable. A student who cannot explain submitted work during the oral defense does not receive credit for that part of the oral assessment.

Contribution statement

Include a table listing each member's main contributions. Every member must understand the full analysis, not only the section they drafted.

Rules for model development

1. Check column names, types, and gross data integrity before splitting. Perform outcome-guided plots, correlations, feature decisions, and model diagnostics on the training data only.
2. Use one reproducible 80/20 train/test split.

3. Compute centering and scaling constants from the training predictors only, and apply the same constants to the test predictors.
4. Create one five-fold `foldid` on the training observations and reuse it for ridge, lasso, elastic net, and every value of `alpha` being compared.
5. Use training-set cross-validation for every tuning and model-selection decision.
6. Keep the test responses locked until all preprocessing, feature maps, tuning rules, and candidate models are fixed.
7. Produce all final holdout predictions together in one final evaluation chunk. Do not revise the analysis after inspecting test performance.

The recommended Gaussian `glmnet` convention is

$$\frac{1}{2n} \|y - Xb\|_2^2 + \lambda \left\{ \frac{1-\alpha}{2} \|b\|_2^2 + \alpha \|b\|_1 \right\}, \quad 0 \leq \alpha \leq 1,$$

with an unpenalized intercept. Ridge has $\alpha = 0$, lasso has $\alpha = 1$, and an interior value gives elastic net (Zou and Hastie 2005; Friedman, Hastie, and Tibshirani 2010).

Problem 1. Prediction design and leakage control (1.00 point)

1A. Target and predictors (0.35)

State the response, candidate predictors, unit of observation, study population, and prediction objective. Cite the assigned dataset source. Record the required exclusions and explain why including an outcome-derived variable would invalidate the comparison.

1B. Split and preprocessing plan (0.35)

Report the split seed, training and test sample sizes, scaling rule, five-fold assignment rule, and missing-value plan. Explain how each step prevents information from the holdout set from entering model construction.

1C. Focused training-data exploration (0.30)

Using only the training observations, report:

- the response distribution;
- one correlation table or focused relationship plot;
- one possible anomaly or influential observation;
- one pair or group of predictors for which multicollinearity may matter.

End with one specific question that the regularized models will help answer.

Problem 2. OLS, ridge, and lasso (3.00 points)

Use the same training observations, standardized design matrix, cross-validation folds, and error definitions throughout this problem. Keep the holdout responses locked.

2A. OLS baseline (0.40)

Fit ordinary least squares. Report the coefficients, training error, five-fold cross-validation error using the shared folds, and one residual or influence diagnostic. If the design is ill-conditioned or coefficients are unstable, explain the evidence instead of removing it from the report.

2B. Ridge regression (0.70)

Fit ridge regression with `alpha = 0`. Report:

- the cross-validation curve;
- `lambda.min` and `lambda.1se`;
- coefficients and training cross-validation error for both choices;
- a coefficient-path plot;
- an explanation of how correlated predictors behave under ridge.

Ridge has no coordinatewise thresholding mechanism and generally does not produce exact zero coefficients. Do not describe a numerically tiny coefficient as an exact selection result.

2C. Lasso regression (0.70)

Fit lasso with `alpha = 1`. Report:

- the cross-validation curve;
- `lambda.min` and `lambda.1se`;
- the nonzero predictors under both choices;
- training cross-validation error;
- a coefficient-path plot;
- an application-specific interpretation of the selected variables.

2D. Controlled comparison and locked core model (0.80)

Construct a training-based comparison table for OLS, ridge, and lasso. Include the tuning rule, cross-validation error, coefficient norm, and number of nonzero slopes. Use this table and an interpretability argument to nominate one **core candidate** before the holdout responses are inspected.

Explain why `lambda.min` and `lambda.1se` answer different decision questions. Distinguish predictive performance from variable selection and interpretability.

2E. Perturbation or stability check (0.40)

Use training data only to investigate one of the following:

- repeat five-fold cross-validation with three prespecified fold seeds;
- duplicate a highly correlated predictor with small random noise;
- bootstrap the training observations and record lasso selection frequencies.

Predict the qualitative behavior of ridge and lasso before running the check, then compare the prediction with the result.

Problem 3. Mathematical mechanisms (1.50 points)

Let $X \in \mathbb{R}^{n \times p}$ be the centered and scaled training design matrix, let $y \in \mathbb{R}^n$ be the centered training response, and define

$$G = \frac{1}{n} X^\top X, \quad z = \frac{1}{n} X^\top y.$$

The intercept is handled separately and is not penalized.

3A. Ridge and conditioning (0.70)

Starting from

$$Q_\lambda(b) = \frac{1}{2n} \|y - Xb\|_2^2 + \frac{\lambda}{2} \|b\|_2^2, \quad \lambda > 0,$$

derive the normal equations and the unique minimizer

$$\hat{b}_\lambda^{\text{ridge}} = (G + \lambda I_p)^{-1} z.$$

Compute the eigenvalues of G from the training design. Report $\kappa_2(G)$, taking it to be infinite when the smallest eigenvalue is zero up to numerical tolerance, and compute

$$\kappa_2(G + \lambda I_p) = \frac{\mu_{\max} + \lambda}{\mu_{\min} + \lambda}$$

at the fitted ridge `lambda.min`. Explain what changed geometrically and numerically.

3B. Lasso optimality and sparsity (0.60)

For

$$R_\lambda(b) = \frac{1}{2n} \|y - Xb\|_2^2 + \lambda \|b\|_1,$$

derive the coordinatewise optimality conditions for a zero and a nonzero coefficient. Then derive the soft-thresholding rule when $G = I_p$:

$$\hat{b}_j = \text{sign}(z_j)(|z_j| - \lambda)_+.$$

Use the conditions to explain why lasso can produce exact zeros while ridge generally does not.

3C. Connect algebra to evidence (0.20)

Choose one result from Problems 2B–2E and explain how it reflects the ridge eigenvalue calculation or the lasso optimality conditions. Refer to a specific coefficient, eigenvalue, path, or selection frequency.

Problem 4. Elastic net and a neural-feature bridge (2.50 points)

This problem is required for every group. You will study an elastic-net model on original predictors and on a fixed nonlinear feature representation. The feature experiment resembles a one-hidden-layer network whose hidden weights are random and frozen; it is **not** a trained deep neural network (Rahimi and Recht 2007; Cho and Saul 2009).

4A. Research question and source map (0.40)

Research one direction connecting regularization to neural networks:

- L_2 penalties and weight decay;
- L_1 penalties and sparse weights;
- unstructured or structured pruning;
- dropout compared with permanent sparsity.

Use at least two primary research papers, including at least one paper not cited in this project brief. Provide a source map with four columns: claim, exact source and locator, what the source establishes, and what it does **not** establish. One source must support a methodological statement, not merely a software instruction.

Useful starting points include weight-decay optimization (Loshchilov and Hutter 2019), dropout (Srivastava et al. 2014), and pruning/compression (Han, Mao, and Dally 2016).

4B. Elastic net on the original predictors (0.50)

Using the shared training folds, fit elastic net for

$$\alpha \in \{0.10, 0.25, 0.50, 0.75, 0.90\}.$$

Tune λ separately for each α , select the pair (α, λ) by training cross-validation, and report the selection rule, coefficient path, nonzero slopes, and cross-validation error. Explain how the fitted solution lies between the ridge and lasso behaviors observed in Problem 2.

4C. Fixed ReLU features (0.70)

Using the standardized training predictors, construct $M = 200$ features

$$h_m(x) = \max\{a_m^\top x + c_m, 0\}, \quad m = 1, \dots, M,$$

where the entries of a_m are generated once from $N(0, 1/p)$ and $c_m \sim N(0, 0.5^2)$. Use one declared seed. Apply the **same** vectors a_m and offsets c_m to the training and test predictors. Remove or handle constant hidden features using training information only.

On this fixed feature matrix, fit ridge, lasso, and elastic net with the shared folds. Report the selected tuning values, training cross-validation errors, and numbers of active hidden features. Explain which parameters were fixed and which were learned.

4D. Lock the analysis and evaluate once (0.50)

Before reading `y_test`, write a short locked-analysis declaration containing:

- the final preprocessing rule;
- the selected OLS/ridge/lasso/elastic-net specifications;
- the random-feature seed and M ;
- the model nominated for deployment;
- the metrics that will be reported.

Then run one final holdout chunk that produces predictions for all fixed candidate models. Report RMSE and MAE in one table. Do not retune, replace predictors, regenerate features, or change the nominated model after viewing this table. Discuss whether the holdout evidence supports the training-based decision.

4E. Deep-learning connection and limitations (0.40)

Explain all of the following in your own words:

1. an L_2 penalty on the trained output weights is a ridge penalty, while weight decay in adaptive neural-network optimization is not always identical to adding an L_2 term;
2. a zero output weight removes one fixed hidden feature from this scalar predictor, but it does not create sparse input-to-hidden weights;
3. dropout randomly masks activations during training and is not the same as permanent pruning;
4. this experiment uses one frozen hidden layer and does not establish a conclusion about a fully trained deep network.

Identify at least two limitations of the experiment, including sensitivity to the random-feature seed. Support external methodological claims with the sources from 4A.

Problem 5. Report quality and reproducibility (1.00 point)

5A. Reproducible submission (0.40)

The submitted archive must render from a clean session. Use relative paths, declared packages, fixed seeds, meaningful object names, a reproduction README, and `sessionInfo()`. The report must show where the holdout responses first enter the computation.

5B. Statistical communication (0.70)

Include an abstract of at most 150 words, a concise comparison table, readable labeled figures, a deployment recommendation, and limitations. Interpret quantities on the correct scale and separate empirical results from mathematical statements.

Problem 6. Individual oral defense (1.00 point)

Each student completes a 6–8 minute oral defense.

Students may be asked to trace one observation through the pipeline, explain why a code change creates leakage, predict the effect of changing λ or the feature seed, or show where a research source

supports a submitted claim.

Point summary

Component	Points
Problem 1: prediction design and leakage control	1.00
Problem 2: OLS, ridge, and lasso	3.00
Problem 3: mathematical mechanisms	1.50
Problem 4: elastic net and neural-feature bridge	2.50
Problem 5: report quality and reproducibility	1.00
Problem 6: individual oral defense	1.00
Total	10.00

References

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